

Automated Vulnerability Assessment Tool

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Abstract — This report documents the design and implementation of an Automated Vulnerability Assessment Tool (AVAT) — a Python-based network security scanner that automates port discovery, service enumeration, vulnerability detection, and PDF report generation. The tool replicates the core reconnaissance workflow used in real-world penetration testing engagements and is made publicly available for educational use via GitHub.

Keywords: Vulnerability Assessment, Nmap, Python, Penetration Testing, Port Scanning, CVE, PDF Report Generation, Network Security

1. INTRODUCTION

Network vulnerability assessment is the process of identifying security weaknesses in a networked environment. It is the first phase of any professional penetration testing engagement. This project automates that process entirely — from scanning a target to generating a structured PDF report — using Python, Nmap, and ReportLab. The tool accepts an IP address or subnet, runs a three-phase Nmap scan, parses all findings, and produces a client-ready PDF report with zero manual steps between phases.

2. TOOL OVERVIEW

2.1 What the Tool Does

1. Accepts a target IP address or subnet range as input.
2. Phase 1 — Fast port discovery: identifies all open ports using Nmap -T4 -F.
3. Phase 2 — Service detection: fingerprints running services and versions (-sV -sC).
4. Phase 3 — Vulnerability scan: runs NSE vulnerability scripts (--script vuln).
5. Parses all findings into a structured data dictionary.
6. Auto-generates a professional PDF report with risk rating and recommendations.

2.2 Technology Stack

Component	Technology	Purpose
Language	Python 3	Core scripting and automation
Network Scanner	Nmap 7.94	Port scanning, service detection, NSE vuln scripts
Python Interface	python-nmap 0.7.1	Programmatic Nmap control from Python
Report Engine	ReportLab 4.x	Automated PDF generation
OS	Kali Linux	Development and testing environment

Table 1: Technology stack used in this project.

3. SYSTEM DESIGN

3.1 File Structure

```
vuln-scanner/ scanner.py # Main scanner and orchestration report_gen.py # PDF report generator requirements.txt # Dependencies README.md #
Setup and usage guide reports/ # Output PDFs (auto-created)
```

3.2 Three-Phase Scan Pipeline

Phase	Nmap Arguments	Output
1 — Discovery	-T4 -F	List of open port numbers
Phase	Nmap Arguments	Output
2 — Enumeration	-sV -sC -p <ports>	Service names and version details
3 — Vulnerability	--script vuln -p <ports>	CVE references and vuln details

Table 2: Three-phase scan pipeline.

3.3 Risk Classification

Risk Level	Condition	Meaning
CRITICAL	Vulnerabilities found > 5	Immediate action required
HIGH	Vulnerabilities found 3 to 5	Urgent remediation needed
MEDIUM	Vulnerabilities found 1 to 2	Scheduled remediation required
LOW	No vulnerabilities detected	Continue monitoring

Table 3: Automated risk classification logic.

4. RESULTS

The tool was tested against localhost (127.0.0.1) on Kali Linux. The following open ports were identified:

Port	Protocol	Service	Finding
80	TCP	HTTP	Web server running — potential web app vulnerabilities
3306	TCP	MySQL	Database exposed — check for default credentials and access controls

Table 4: Scan results from test run on localhost (127.0.0.1).

A PDF report was automatically generated upon scan completion, containing host details, the port and service table, vulnerability findings, risk level, and a recommendations section — requiring no manual steps from the user.

5. HOW TO USE THIS TOOL

5.1 Requirements

- Kali Linux (recommended) or any Linux distribution with Nmap installed.
- Python 3.6 or above.
- Nmap installed: `sudo apt install nmap`
- Python dependencies: `pip3 install python-nmap reportlab --break-system-packages`

5.2 Installation

```
git clone https://github.com/SYEDMAHAMMEDAFZAL/vuln-scanner.git cd vuln-scanner pip3 install -r requirements.txt --break-system-packages
```

5.3 Usage

```
# Scan a single host sudo python3 scanner.py 192.168.1.1 # Scan a subnet sudo python3 scanner.py 192.168.1.0/24 # PDF report saved
automatically to reports/ folder
```

6. CONCLUSION

The Automated Vulnerability Assessment Tool successfully automates the core penetration testing reconnaissance workflow in approximately 300 lines of Python. It demonstrates practical competency in network security, Python scripting, Linux systems, and security documentation. The tool is open-source and freely available for educational and authorised professional use via GitHub.

Legal Notice: This tool must only be used against systems you own or have explicit written permission to test. Unauthorised network scanning is illegal under the IT Act 2000 (India), CFAA (USA), and equivalent legislation in other jurisdictions. The author bears no responsibility for misuse of this tool.

```
(kalikali@vbox)-[~/vuln-scanner]  
$ sudo python3 scanner.py
```

VULN SCAN

Automated Vulnerability Assessment Tool | by Afzal

```
[!] WARNING: Only scan systems you own or have permission to test.  
[!] Unauthorized scanning is illegal.
```